ROB 345/549
Assignment 1

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Problem 1

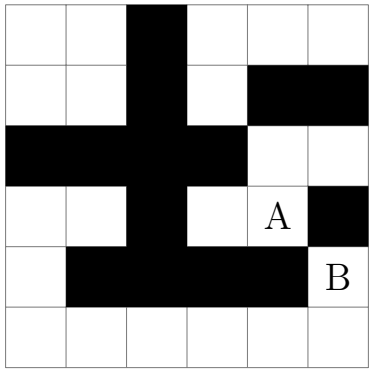
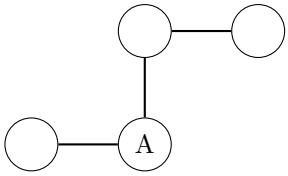


Figure 1: Environment for Problem 1.

(a)



(b)

1. **Iteration 1**: $Q = \{A\} = \{(5, 3)\}$;
2. $x = Q$.getVertex() $= (5, 3)$;
3. Neighbours of x: {(5, 4), (4, 3)};
4. Mark these as visited and add to Q.;
   // Note we put $(5, 3)$ into the queue first as this vertex has a child vertext $(6, 4)$ while $(4, 3)$ has not, and as a result if we add $(4, 3)$ first, the following precedure will be exactly the same as the DFS.
5. **Iteration 2**: $Q = \{(5, 4), (4, 3)\}$;
6. $x =$ Q.getVertex() $= (5, 4)$;
7. Neighbours of x: $(6, 4)$;
8. Mark it as visited and add to Q;
9. **Iteration 3**: $Q = \{(4, 3), (6, 4)\}$;
10. $x = Q$.getVertex() $= (4, 3)$;
11. Neighbours of x: $\varnothing$;
12. **Iteration 4**: $Q = \{(6, 4)\}$;
13. $x = Q$.getVertex() $= (6, 4)$;
14. Neighbours of x: $\varnothing$;
15. **Iteration 5**: $Q = \varnothing$, Terminate;

(c)

1. **Iteration 1**: $Q = \{A\} = \{(5, 3)\}$;
2. $x = Q$.getVertex() $= (5, 3)$;
3. Neighbours of x: {(5, 4), (4, 3)};

4. Mark these as visited and add to Q;
   // Note we now put the vertices at the beginning of the current queue (stack), LIFO.
5. **Iteration 2**: $Q = \{(5,4),(4,3)\}$;
6. $x = \text{Q.getVertex}() = (5,4)$
7. Neighbours of x: $(6,4)$
8. Mark it as visited and add to Q;
9. **Iteration 3**: $Q = \{(6,4),(4,3)\}$;
10. $x = Q.\text{getVertex}() = (6,4)$;
11. Neighbours of x: $\varnothing$;
12. **Iteration 4**: $Q = \{(4,3)\}$
13. $x = Q.\text{getVertex}() = (4,3)$
14. Neighbours of x: $\varnothing$
15. **Iteration 5**: $Q = \varnothing$. Terminate

Problem 2

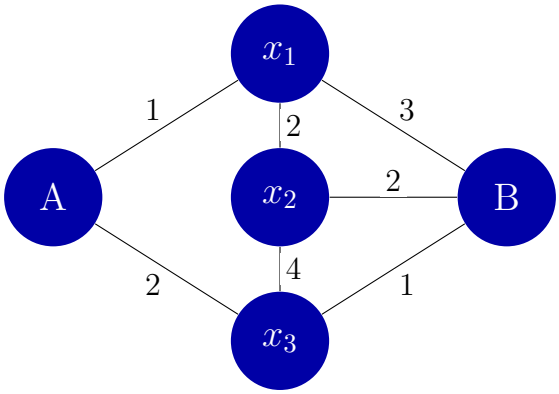


Figure 2: Graph for Problem 2.

(a) Let $H(A) = 5$, $H(x_1) = 3$, $H(x_2) = 2$, $H(x_3) = 4$, and $H(B) = 0$. Then $F(A) = C(A) + H(A) = 0 + 5 = 5$. Hence $Q = \{A : 5\}$

Iteration 1:

1. $x = A$
2. Neighbours: $\{x_1, x_3\}$
   $C(x_1) = 1$, $F(x_1) = 1 + 3 = 4$
   $C(x_3) = 2$, $F(x_3) = 2 + 4 = 6$
3. Dead vertex: $\{A\}$
4. $Q = \{x_1 : 4, x_3 : 6\}$

Iteration 2:

1. $x = x_1$ for lowest F
2. Neighbours: $\{x_2, B\}$
   $C(x_2) = 1 + 2 = 3$, $F(x_2) = 2 + 3 = 5$
   $C(B) = 1 + 3 = 4, F(B) = 4 + 0 = 4$
3. Dead vertex: $\{A, x_1\}$
4. $Q = \{B : 4, x_2 : 5, x_3 : 6\}$

Iteration 3:

1. $x = B$ for lowest F. Seceed.

(b) The cost of the path $A \to x_3 \to B$ is 3, which is lower than that of the path found in (a). To find the optimal path, we can set the heuristic function to 0 for all vertices. This reduces A* to Dijkstra's algorithm, which guarantees an optimal path when all edge costs are non-negative.

Iteration 1:

1. $x = A$
2. Neighbours: $\{x_1, x_3\}$
   $C(x_1) = 1$, $C(x_3) = 2$
3. Dead vertex: $\{A\}$
4. $Q = \{x_1 : 1, x_3 : 2\}$

Iteration 2:

1. $x = x_1$ for minimum cost
2. Neighbours: $\{x_2, B\}$
   $C(x_2) = 1 + 2 = 3$, $C(x_B) = 1 + 3 = 4$
3. Dead vertex: $\{A, x_1\}$
4. $Q = \{x_3 : 2, x_2 : 3, B : 4\}$

Iteration 3:

1. $x = x_3$ for minimum cost
2. Neighbours: $\{x_2, B\}$
   $C(x_2) = 2 + 4 = 6$, $C(B) = 2 + 1 = 3$
3. Dead vertex: $\{A, x_1, x_3\}$
4. $Q = \{x_2 : 3, B : 3, B : 4\}$

Iteration 4:

1. $x = x_2$ for minimum cost
2. Neighbours: $\{x_3, B\}$
   $C(x_3) = 3 + 4 = 7$, $C(B) = 3 + 2 = 5$
3. Dead vertex: $\{A, x_1, x_3, x_2\}$
4. $Q = \{B : 3, B : 4, B : 5, x_3 : 7\}$

Iteration 5:

1. $x = B$ with $C(B) = 3$. Successfully find the optimal path.